
The Proxy Romberg Ratio (pRR) in Repeated-Trial Stabilometry: Estimator Choice, Reliability, and Latent Sensory Structure

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Abstract

The Romberg ratio classically contrasts eyes-closed (EC) and eyes-open (EO) postural sway. Many stabilometric datasets, however, contain repeated trials under a single condition with no EO/EC annotation, yet ratio-type contrast scores are still computed. Such scores are mathematically ambiguous under repetition (ratio of means vs. mean of paired ratios vs. robust/bootstrap estimators), and their stability depends on the underlying feature. We study a proxy Romberg ratio (pRR) on repeated center-of-pressure (CoP) trials, including an energetics-inspired index derived from CoP kinematics and body mass. Using an internal dataset of healthy young adults ($N = 80$ with ≥ 10 free-condition trials; $f_s = 250$ Hz; 30 s trials), we compute 24 stabilometric measures per trial, build subject-specific low- vs. high-sway proxy groups by a median split, and compare four pRR estimators: ratio of group means (A), mean of paired ratios (B), bootstrap median (C), and geometric mean of paired ratios (D_{geo}). Agreement is assessed by Bland–Altman analysis on $\ln(\text{pRR})$; internal consistency and dimensionality by Cronbach’s α and PCA; and subject profiles by k -means clustering with cross-estimator stability quantified by the adjusted Rand index (ARI). Across five split-features, D_{geo} most closely tracks A ($\text{MSE}_{\ln} \approx 0.011\text{--}0.012$), while

the bootstrap median C is roughly twice as dispersed. The curated five-measure battery is near-unidimensional (PC1 explained variance $\approx 0.96\text{--}0.98$; Cronbach’s $\alpha \approx 0.96$) and yields two robust subject profiles ($k = 2$) recovered consistently across estimators and split-features (ARI up to 1.0). A trial-level reliability analysis shows that a single ratio is only fairly reliable ($\text{ICC} \approx 0.37\text{--}0.43$) but the 5-trial aggregate is good (Cronbach’s $\alpha \approx 0.76\text{--}0.80$), with a two-facet generalizability coefficient $E\rho^2 = 0.76$. We recommend D_{geo} as a robust default and stress explicit reporting of the estimator and proxy-labeling choices.

Keywords: stabilometry; center of pressure; proxy Romberg ratio; repeated measures; bootstrap; Bland–Altman; Cronbach’s alpha; principal component analysis; clustering; energy-expenditure index.

Introduction

The Romberg test assesses postural control by comparing balance with eyes open (EO) and eyes closed (EC). In quantitative posturography, Romberg-type indices are commonly expressed as ratios between EC and EO values of a center-of-pressure (CoP) sway metric such as path length, mean velocity, or sway area (Prieto et al., 1996). Many practical datasets, however, consist of repeated trials acquired under a single experimental condition with no explicit EO/EC trial annotation. Re-

searchers nonetheless frequently compute ratio-based contrast scores between subsets of trials to summarize within-subject variation and to define individual balance phenotypes.

Repeated-trial ratio scoring is fundamentally ambiguous. The ratio of group means, $\bar{X}_{\text{high}}/\bar{X}_{\text{low}}$, is in general not equal to the mean of paired trialwise ratios,

$$\frac{1}{n} \sum_{i=1}^n \frac{X_{\text{high},i}}{X_{\text{low},i}}, \quad (1)$$

and robust estimators (for example, bootstrap medians) target yet other functionals of the underlying ratio distribution (Efron and Tibshirani, 1993). Because ratio distributions are typically right-skewed, log-scale aggregation and geometric averaging are often more stable and interpretable. Beyond the estimator, stability also depends on the chosen feature: some measures yield tight trial-to-trial behaviour while others are noisy.

The construct that motivates Romberg-type scoring is the individual sensory balance: the relative reliance of a person’s postural control on vision versus the proprioceptive and vestibular channels. In the classic paradigm this is probed by removing vision; a large EC/EO ratio is read as heavy visual dependence. When only repeated single-condition trials are available, one cannot recover that physiological contrast, but one can still ask a related, well-posed question: does a stable, low-dimensional individual signature emerge from the within-condition variability of sway, and is that signature reproducible across the analytic choices used to compute it? A robust affirmative would indicate that repeated-trial contrast scores capture a genuine person-level property—a candidate sensory-balance proxy—even without explicit visual manipulation, while a negative result would caution against over-interpreting such scores.

Two further considerations motivate a careful treatment. The first is reliability: when a single contrast score is to be used as an individual phenotype, it must be internally consistent across the measures that compose it and reproducible across reasonable analytic choices. Posturographic measures are known to be intercorrelated (Prieto et al., 1996), so a battery of ratios may carry far less independent information than its dimensionality suggests. The second is sensitivity to the labelling rule itself: in the absence of EO/EC annotation, the partition into “low” and “high” trials is an analyst choice, and a contrast score is only trustworthy if its structure survives plausible variation in that choice. We therefore treat the split-feature as an explicit sensitivity

axis rather than a fixed convention.

This paper has two aims. First, we provide a mathematically explicit framework for comparing repeated-trial ratio estimators, and we ask whether stable low-dimensional structure and reproducible subject profiles can be recovered from the resulting proxy Romberg ratio (pRR). We emphasise throughout that the proxy labelling does not correspond to visual deprivation, so pRR must not be read as a physiological Romberg ratio. Second, we incorporate an energetics-inspired index derived from CoP kinematics and body mass, computed according to the integral posture-regulation estimate of Grokhovsky and Kubryak (Grokhovsky and Kubryak, 2018), and assess its reliability and contribution alongside the conventional kinematic measures.

Methodology

Study design and data availability

Data originate from an internal study of healthy young adults and are not publicly available owing to privacy and institutional constraints. Derived feature tables and analysis scripts can be shared on reasonable request, subject to approval. Sex distribution, stabiloplatform manufacturer/model, ethical approval, informed-consent procedure, and funding are documented separately and are marked [TO FILL] in the declarations below.

Database structure and data sources

The dataset is stored as a combination of SQLite tables and a Parquet time-series file (Figure 1): `subjects` holds subject-level metadata (`subject_id`, age, height, mass in kg, reconstructed as a per-subject average during cleaning); `metadata_cleaned` holds trial metadata keyed by `file_id` (`subject_id`, `condition`, `trial`, `duration_samples`); `signals_cleaned.parquet` holds the cleaned CoP time series keyed by `file_id` (`sample_num`, `cop_x`, `cop_y` in millimetres); and `segment_labels` holds machine-generated within-trial annotations from an unrelated task that are not used here.

Cohort and trial inclusion

All analyses are restricted to trials with `condition="free"`, the only condition with enough repetitions per subject for paired ratio scoring (the `stress` and `euro` conditions contain

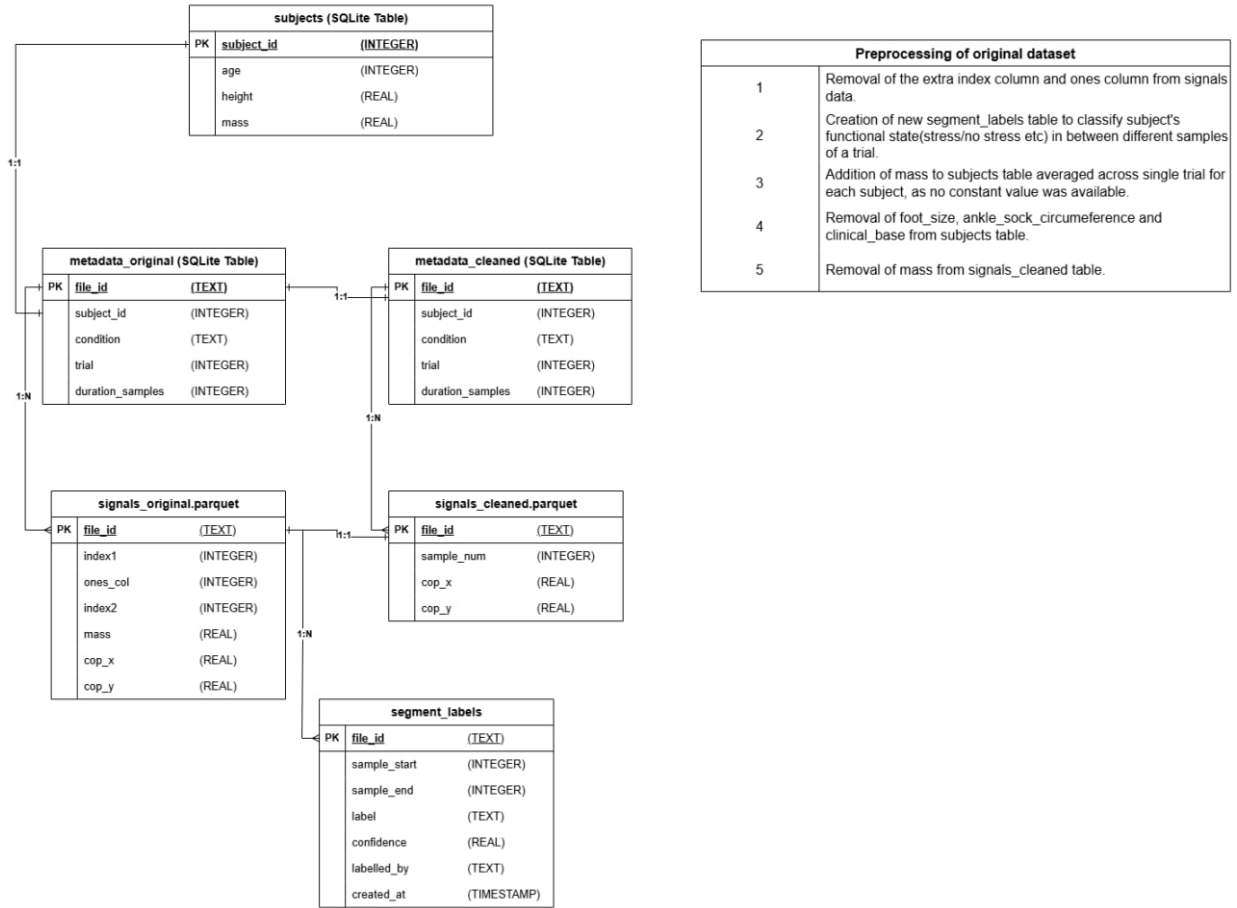


Figure 1: Entity-relationship schema and preprocessing log of the source dataset. The present analysis joins `signals_cleaned.parquet` to `metadata_cleaned` and `subjects` by `file_id` and `subject_id`; the `segment_labels` table and the original (pre-cleaning) signal columns are not used.

far fewer trials). The inclusion funnel is as follows: of the 124 subjects in the database, 85 had at least one free-condition trial; we further required at least ten free trials per subject so that a balanced five-versus-five proxy partition could be formed, which leaves a working cohort of $N = 80$ subjects. The five excluded subjects had too few free trials to support the design (one with two trials, four with nine). The retained cohort is predominantly young adults (mean age 20.4 ± 5.1 years; height 170.0 ± 9.7 cm; mass 67.7 ± 16.1 kg), although a small number of metadata entries are clearly anomalous (for example, an implausible recorded age of two years); these do not affect the within-subject ratio analysis, in which per-subject scale cancels and age/height are used only descriptively. Each retained subject contributed on average 11.2 free trials; trials are 30 s long (≈ 7500 samples at 250 Hz).

Signal handling

We deliberately use the cleaned CoP signals as provided, applying only the transformations the feature pipeline requires: coordinates are converted from millimetres to metres, and each axis is mean-centred per trial, $x_c[n] = x[n] - \bar{x}$ and $y_c[n] = y[n] - \bar{y}$. We do not impose additional low-pass filtering or outlier suppression on top of the existing cleaning, so that the reported measures reflect the cleaned signals directly. Because pRR is a within-subject ratio of features, any monotone per-subject scale introduced by such preprocessing would in any case largely cancel.

Stabilometric measures

For each trial we compute 24 measures spanning amplitude, geometry, velocity, frequency, complexity, and energetics. Let $X[n], Y[n]$ ($n = 1, \dots, N$) denote centred CoP in metres, $\Delta t = 1/f_s$, $T = (N - 1)\Delta t$, and write per-step increments $\Delta X_n = X[n+1] - X[n]$, $\Delta Y_n = Y[n+1] - Y[n]$. Representative definitions

are

$$L = \sum_{n=1}^{N-1} \sqrt{\Delta X_n^2 + \Delta Y_n^2}, \quad \bar{v} = L/T, \quad (2)$$

$$\text{RMS}_{\text{tot}} = \sqrt{\text{RMS}_X^2 + \text{RMS}_Y^2}, \quad (3)$$

$$\text{Area}_{95} = \pi \chi_{2,0.95}^2 \sqrt{\det \mathbf{C}}, \quad (4)$$

where $\mathbf{C}$ is the 2×2 covariance of $[X, Y]$ and $\chi_{2,0.95}^2 \approx 5.991$ (Prieto et al., 1996). Speed is $v[n] = \sqrt{V_X[n]^2 + V_Y[n]^2}$ with $V_X[n] = \Delta X_n / \Delta t$; from it we derive mean, 95th-percentile, coefficient-of-variation, and peak speed, mean/peak acceleration, spectral centroid and Welch median frequency, and approximate entropy. The battery also includes per-axis RMS, ranges, robust ($P_{95} - P_5$) ranges, velocity zero-crossing rates, and the energetic quantities defined next.

Energy-expenditure index

We use an energetics-inspired index based on increments of squared CoP velocity weighted by body mass m (kg), computed according to Grokhovsky and Kubryak (Grokhovsky and Kubryak, 2018). Writing the combined squared speed $U[n] = V_X^2[n] + V_Y^2[n]$, the source defines the per-step energy contribution by combining the two axes and then taking the magnitude of the increment:

$$A[n] = \frac{m}{2} |U[n+1] - U[n]|. \quad (5)$$

The total index, average power, and energy-per-metre index are

$$A = \sum_n A[n], \quad P = \frac{A}{T}, \quad V_{\text{exp}} = \frac{A}{L + \varepsilon}, \quad (6)$$

with a small ε guarding the division. A signed increment telescopes to a trivial endpoint difference, so the magnitude form in Equation (5) is required for a meaningful path-resolved energy proxy.

Proxy labelling of low/high-sway groups

Because EO/EC labels are unavailable, we partition each subject's free trials into a low-sway and a high-sway group by a median split on a chosen split-feature S . For subject s with trials $j = 1, \dots, J_s$ and $\text{med}_s = \text{median}_j(S_{s,j})$, trials with $S_{s,j} \leq \text{med}_s$ form the LOW group and those with $S_{s,j} > \text{med}_s$ the HIGH group; each group is ordered by trial index and trimmed to five trials, yielding a balanced five-pair design. The split-feature is a geometric/sway quantity, so the partition reflects postural amplitude rather than any single derived index.

Proxy Romberg ratio and estimators

For any feature X the proxy Romberg ratio is $\text{pRR} = X_{\text{HIGH}}/X_{\text{LOW}}$. With paired trials ($i = 1, \dots, 5$) and trialwise ratios $R_i = X_{\text{HIGH},i}/X_{\text{LOW},i}$, we evaluate four estimators,

$$\text{pRR}_A = \bar{X}_{\text{HIGH}}/\bar{X}_{\text{LOW}}, \quad (7)$$

$$\text{pRR}_B = \frac{1}{5} \sum_{i=1}^5 R_i, \quad (8)$$

$$\text{pRR}_C = \text{median}\{R_i\} \text{ (bootstrap)}, \quad (9)$$

$$\text{pRR}_{D_{\text{geo}}} = \exp\left(\frac{1}{5} \sum_i \ln R_i\right), \quad (10)$$

namely the ratio of group means (A), the mean of paired ratios (B), the bootstrap median of the paired ratios (C, from $B = 2000$ resamples), and the geometric mean of paired ratios (D_{geo}). Estimator A serves as the conventional reference because it is the form most often reported in posturography.

Agreement, consistency, structure, clustering

Agreement between estimators is quantified by Bland–Altman analysis (Bland and Altman, 1986) on $\ln(\text{pRR})$, reporting bias, $\text{SD}(\Delta)$, limits of agreement bias $\pm 1.96 \text{SD}(\Delta)$, and the mean squared log-difference $\text{MSE}_{\ln}$ against A. Working on the log scale converts the multiplicative structure of a ratio into an additive one, symmetrises the right-skewed distribution, and makes a difference of $\ln(\text{pRR})$ interpretable as a percentage discrepancy. Internal consistency of the subject $\times$ measure matrix is summarised by Cronbach's α (Cronbach, 1951),

$$\alpha = \frac{p}{p-1} \left(1 - \frac{\sum_{m=1}^p \sigma_m^2}{\sigma_{\text{tot}}^2}\right), \quad (11)$$

with p measures (“items”), per-measure variances σ_m^2 across subjects, and total variance σ_{tot}^2 of the summed profile. Because the five trials provide repeated observations, α admits the reliability reading of an intraclass correlation in the sense of Shrout and Fleiss (Shrout and Fleiss, 1979). Dimensionality is assessed by PCA on z-scored $\ln(\text{pRR})$ (Jolliffe and Cadima, 2016), reporting first-component explained variance (PC1) and loadings; we also fit a single-factor model and report its root-mean-square reconstruction error as a direct check of unidimensionality. Subjects are clustered with k -means in a curated five-measure battery (`mean_speed_mps`, `path_length_m`, `mean_accel_mps2`, `energy_A`, `speed_p95_mps`); k is chosen by maximising the average silhouette on $\ln(\text{pRR}_{D_{\text{geo}}})$, and cross-estimator stability is measured by the adjusted Rand index (ARI) rela-

tive to D_{geo} , where $\text{ARI} = 1$ denotes identical clusterings and 0 chance agreement. We repeat the whole pipeline for five split-features (`ellipse_area95_m2` [primary], `mean_speed_mps`, `path_length_m`, `rms_total_m`, `speed_p95_mps`) to test robustness to the proxy-labelling choice.

Reliability and generalizability

We assess the reliability of the proxy ratio at the trial level using the five paired trialwise ratios R_i ($i = 1, \dots, 5$) on $\ln$ scale. For each measure we compute the single-measurement intraclass correlation $\text{ICC}(2, 1)$ from a two-way random-effects model (subjects $\times$ trials) (Shrout and Fleiss, 1979),

$$\text{ICC}(2, 1) = \frac{\text{MS}_R - \text{MS}_E}{\text{MS}_R + (k-1)\text{MS}_E + \frac{k}{n}(\text{MS}_C - \text{MS}_E)}, \quad (12)$$

with $k = 5$ trials, n subjects, and row/column/error mean squares $\text{MS}_R, \text{MS}_C, \text{MS}_E$. Cronbach's α over the five trials equals the reliability of the 5-trial mean, $\text{ICC}(3, k) = (\text{MS}_R - \text{MS}_E)/\text{MS}_R$. The standard error of measurement is $\text{SEM} = \sqrt{\text{MS}_E}$ and the smallest detectable change is $\text{SDC} = 1.96\sqrt{2}\text{SEM}$ (both in $\ln$ units, i.e. proportional). For variant D we report the mean within-subject coefficient of variation of R_i . Following the construct in the task brief we also compute a contrast fraction $\text{VC} = 1 - 1/R$ (the proxy analogue of a “visual contribution”; because the labelling is sway-based and not EO/EC, this quantifies the high-versus-low contrast magnitude, *not* visual dependence). Finally, we run a two-facet generalizability study (Brennan, 2001) on the balanced subjects $\times$ measures $\times$ trials array of $\ln R$, decomposing the variance into subject, measure, trial, and interaction components by random-effects ANOVA and forming the generalizability coefficient $E\rho^2 = \sigma_p^2/(\sigma_p^2 + \sigma_\delta^2)$ with $\sigma_\delta^2 = \sigma_{pm}^2/n_m + \sigma_{pt}^2/n_t + \sigma_{pmt}^2/(n_m n_t)$ for a design that averages over $n_m = 5$ measures and $n_t = 5$ trials.

Results

Estimator agreement

On the logarithmic scale, the geometric mean of paired ratios D_{geo} tracks the conventional ratio-of-means estimator A closely (bias near zero, $\text{SD}(\Delta) \approx 0.11$), whereas the bootstrap median C is roughly twice as dispersed; the corresponding Bland–Altman comparison for the primary

split-feature is shown in Figure 2. Across all five split-features (Table 1), D_{geo} gives the smallest log-scale disagreement with A among the nontrivial estimators ($\text{MSE}_{\ln} \approx 0.011\text{--}0.012$, $\text{SD}(\Delta) \approx 0.103\text{--}0.110$); the mean-of-ratios estimator B is intermediate ($\text{MSE}_{\ln} \approx 0.004\text{--}0.006$) and C is the most dispersed ($\text{SD}(\Delta) \approx 0.19\text{--}0.21$). The ordering $B < D_{\text{geo}} < C$ in raw MSE reflects that B and A are both arithmetic-mean constructs, whereas D_{geo} , although geometric, stays close to A while resisting the heavy-tailed instability that inflates C.

A near-unidimensional curated battery

PCA on z-scored $\ln(\text{pRR}_{D_{\text{geo}}})$ (Figure 3) shows that the full 24-measure set is dominated by a single component but is not unidimensional (PC1 explained variance $\approx 0.56\text{--}0.61$), whereas the curated five-measure battery is near-unidimensional (PC1 $\approx 0.96\text{--}0.98$). The five curated measures load almost equally and positively on PC1 (Figure 6, left), so PC1 acts as a single “global pRR magnitude” axis. A one-factor latent model fitted to the curated battery reconstructs the standardized $\ln(\text{pRR})$ with $\text{RMSE} \approx 0.20$ (vs. ≈ 0.67 for the full 24-measure set), corroborating the strong one-dimensional structure of the curated battery.

Internal consistency

Cronbach's α for $\ln(\text{pRR}_{D_{\text{geo}}})$ is high for both the full battery ($\alpha \approx 0.946\text{--}0.951$) and the curated battery ($\alpha \approx 0.960\text{--}0.965$), indicating that the subject-level pRR profile is internally consistent across measures rather than dominated by a single noisy feature.

Inter-feature agreement

To examine how the contrast score behaves across the measures that compose it, we form the correlation matrix of $\ln(\text{pRR}_{D_{\text{geo}}})$ across subjects for the curated battery (Figure 4). All pairwise correlations are strongly positive (Pearson $r = 0.70\text{--}1.00$; mean 0.87), confirming that a subject who shows a large high-versus-low contrast on one measure tends to show it on the others. The velocity-derived measures (mean speed, path length, 95th speed percentile) are mutually near-collinear ($r \geq 0.99$), expected because over fixed-duration trials path length is essentially mean speed integrated over time. The energy-expenditure index is also tightly aligned with the kinematic measures ($r = 0.82\text{--}0.91$), so it reinforces rather than diversifies the shared contrast dimension; mean acceleration is the least redundant member

Bland-Altman agreement of pRR estimators against A (ratio of group means)

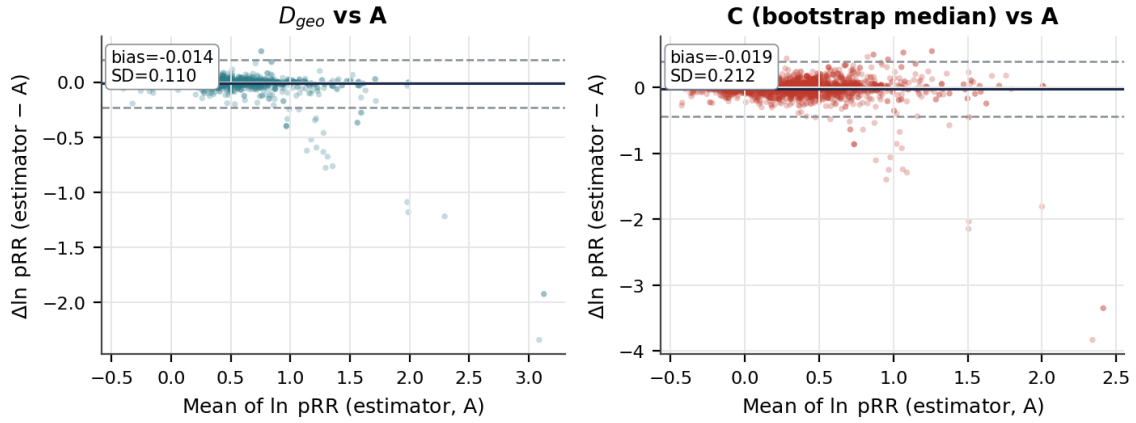


Figure 2: Bland-Altman agreement of pRR estimators against estimator A (ratio of group means) on the $\ln(\text{pRR})$ scale, pooled over subjects and measures for the primary split-feature (`ellipse_area95_m2`). Solid line: mean difference; dashed lines: ± 1.96 SD limits of agreement. D_{geo} (left) is tightly centred on A; the bootstrap median C (right) is markedly more dispersed.

Table 1: Estimator agreement against A on $\ln(\text{pRR})$, across split-features ($N = 80$, 24 measures). $\text{MSE}_{\ln}$ is the mean squared log-difference; SD_{Δ} is the Bland-Altman dispersion.

Split feature	n_S	$\text{MSE}_{\ln}(\text{B})$	$\text{SD}_{\Delta}(\text{B})$	$\text{MSE}_{\ln}(\text{C})$	$\text{SD}_{\Delta}(\text{C})$	$\text{MSE}_{\ln}(D_{\text{geo}})$	$\text{SD}_{\Delta}(D_{\text{geo}})$
<code>ellipse_area95_m2</code>	80	0.0041	0.0529	0.0454	0.2124	0.0123	0.1100
<code>mean_speed_mps</code>	80	0.0048	0.0586	0.0398	0.1987	0.0113	0.1056
<code>path_length_m</code>	80	0.0048	0.0586	0.0398	0.1987	0.0113	0.1056
<code>rms_total_m</code>	80	0.0061	0.0664	0.0442	0.2098	0.0122	0.1094
<code>speed_p95_mps</code>	80	0.0051	0.0599	0.0370	0.1917	0.0107	0.1029

($r \approx 0.70$). This is the correlational counterpart of the near-unidimensional PCA result.

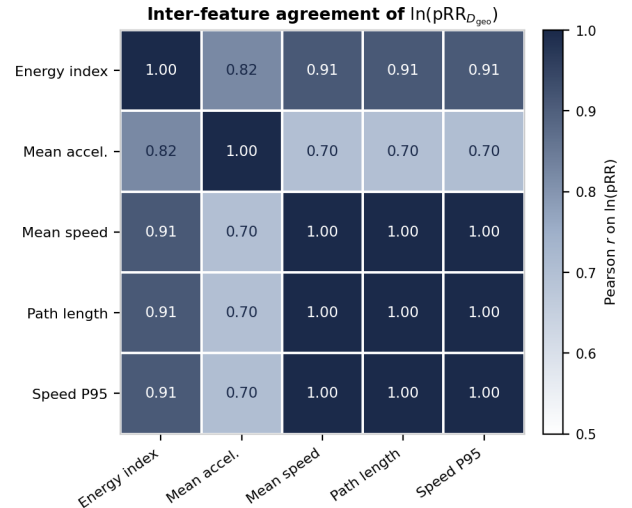


Figure 4: Pearson correlation matrix of $\ln(\text{pRR}_{D_{\text{geo}}})$ across subjects for the five curated measures (primary split-feature, $N = 80$). Off-diagonal correlations range from 0.70 to 1.00 (mean 0.87); the energy index aligns with the kinematic measures ($r = 0.82$ –0.91).

Dimensionality of the subject-level pRR profile (log scale)

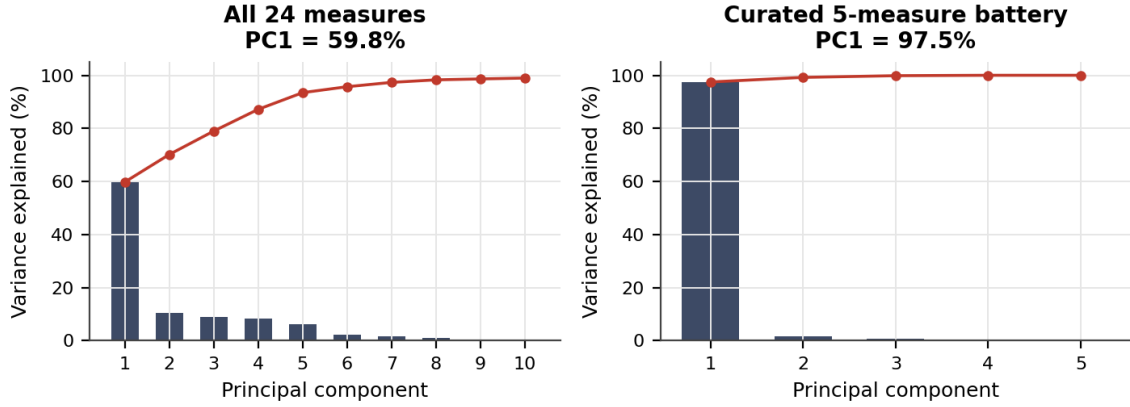


Figure 3: Scree plots (bars: per-component variance; line: cumulative) of PCA on z-scored $\ln(\text{pRR}_{D_{\text{geo}}})$ for all 24 measures (left) and the curated 5-measure battery (right). The curated battery is near-unidimensional.

Reliability, stability, and generalizability

Although the curated battery is internally consistent *across measures*, the reliability of a *single trialwise* ratio is only fair: ICC(2,1) for the five curated measures ranges 0.37–0.43 (Table 2, Figure 5b), meaning that one high-versus-low pairing is dominated by trial-to-trial variability. Aggregating the five trials restores good reliability—Cronbach’s α over the five trials is 0.76–0.80, the reliability of the 5-trial mean—which is the quantitative justification for using a multi-trial estimator rather than a single ratio. Among measures, the velocity-derived ones (mean speed, path length) are the most stable (lowest within-subject CV ≈ 0.24 , smallest SDC), whereas the energy index is the noisiest (CV ≈ 0.47 , SDC_{ln} ≈ 1.57); the trialwise-ratio distributions in Figure 5a show the same ordering. The proxy contrast fraction VC = $1 - 1/R$ averages 0.34 across subjects (median 0.34; 94.5% of subject–measure ratios exceed unity), and is largest for the energy index (0.48); we stress that this is a contrast-magnitude descriptor, not a measure of visual dependence.

The two-facet generalizability study (Figure 5c) decomposes the variance of $\ln R$ into subject (29%), measure (13%), trial (4%), subject $\times$ trial (39%), subject $\times$ measure (5%), measure $\times$ trial (0.4%), and residual (11%) components. The dominant subject $\times$ trial term confirms that within-subject trial variability—not between-measure disagreement—is the principal noise source, exactly the pattern that depresses single-trial ICC while leaving the multi-measure profile consistent. For a design that averages over five measures and five trials the generalizability coefficient is $E\rho^2 = 0.76$, i.e. acceptable depend-

ability is achievable but contingent on aggregating repeated trials.

Table 2: Trial-level reliability of the proxy Romberg ratio for the curated measures (primary split, $N = 80$, balanced 5×5 design, on $\ln R$). ICC(2,1): single-trial reliability; α_5 : Cronbach’s alpha of the 5-trial mean; SEM, SDC in $\ln$ units; CV: mean within-subject coefficient of variation (variant D); VC: mean contrast fraction $1 - 1/R$.

Measure	ICC(2,1)	α_5	SEM _{ln}	SDC _{ln}	CV	VC
Mean speed	0.43	0.80	0.26	0.71	0.24	0.31
Path length	0.43	0.80	0.26	0.71	0.24	0.31
Mean accel.	0.40	0.78	0.24	0.66	0.22	0.26
Energy index	0.37	0.76	0.57	1.57	0.47	0.48
Speed P95	0.37	0.76	0.29	0.82	0.27	0.32

Two robust subject profiles

The optimal cluster number is $k = 2$ for every split-feature, with moderate silhouette values (≈ 0.50 – 0.58). The two profiles separate cleanly along PC1 (Figure 6, right), consistent with a low-pRR and a high-pRR phenotype: one group shows a markedly larger high-versus-low-sway contrast across the whole curated battery while the other shows a compressed contrast, and because PC1 carries almost all of the curated-battery variance this single axis is what the clustering effectively thresholds. Cross-estimator stability against D_{geo} is strong for A and B (ARI ≈ 0.81 – 0.85 for A; ≈ 0.90 – 1.00 for B) but weaker for the bootstrap median C (ARI ≈ 0.52 – 0.72), echoing C’s larger Bland–Altman dispersion: estimators that

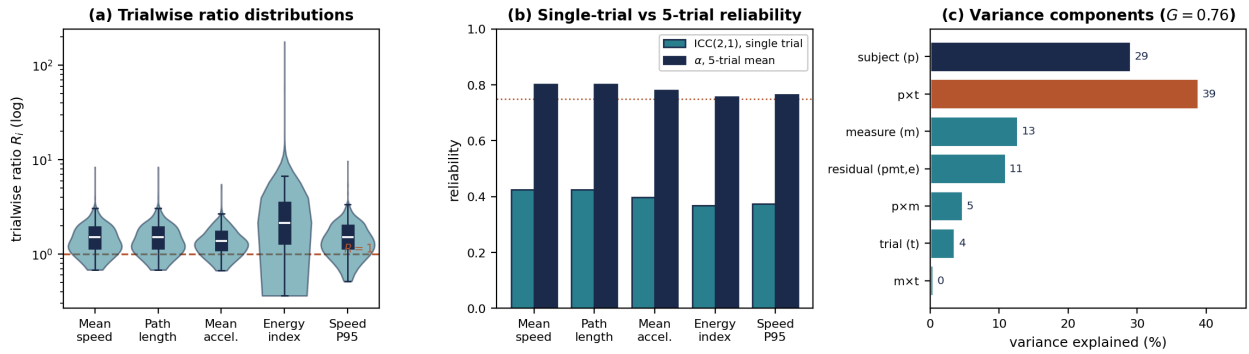


Figure 5: Reliability and generalizability of the proxy Romberg ratio (primary split, $N = 80$). (a) Trialwise-ratio R_i distributions per curated measure (log scale; dashed line $R = 1$); the energy index is the most dispersed. (b) Single-trial ICC(2,1) (fair, ≈ 0.37 – 0.43) versus the 5-trial Cronbach α (good, ≈ 0.76 – 0.80); dotted line at 0.75. (c) Generalizability-theory variance components; the subject $\times$ trial interaction dominates, and $E\rho^2 = 0.76$ for averaging over five measures and five trials.

agree on the value of the ratio also agree on which subjects fall in which profile. That the same two-profile partition re-emerges across all five split-features is the strongest single piece of evidence that the structure reflects a genuine person-level property rather than an artefact of one labelling rule. Table 3 summarises consistency, dimensionality, and clustering across split-features.

Representative centre-of-pressure trajectories (subject 48)

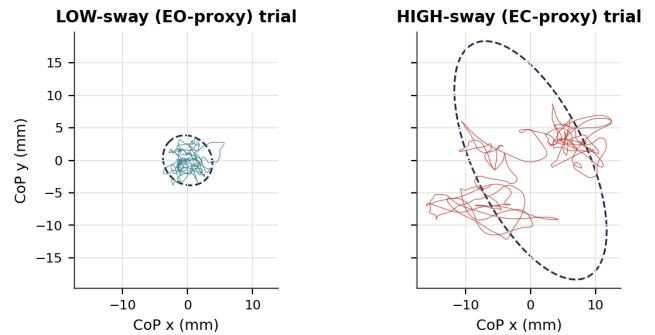


Figure 7: Representative centred CoP trajectories (single subject) for a LOW-sway (EO-proxy) and a HIGH-sway (EC-proxy) trial, each with its 95% confidence ellipse (dashed). The proxy labelling reflects sway magnitude under a single condition, not visual deprivation.

Discussion

Representative trajectories

Figure 7 illustrates the proxy partition with a low-sway and a high-sway trial from the same subject, each overlaid with its 95% confidence ellipse, making concrete the geometric contrast that the median split encodes.

Two findings stand out. First, repeated-trial ratio scoring is genuinely estimator-dependent, and the choice has measurable consequences for agreement and for the stability of any downstream profiling. Among nontrivial aggregators, the geometric mean of paired ratios (D_{geo}) most closely tracks the conventional ratio-of-means (A) while resisting the heavy-tailed instability that inflates the bootstrap median (C). Because ratio distributions are right-skewed, this is expected: geometric averaging is the natural location estimator on the log scale, where the analysis is in any case best conducted.

Latent structure of subject pRR profiles (curated battery)

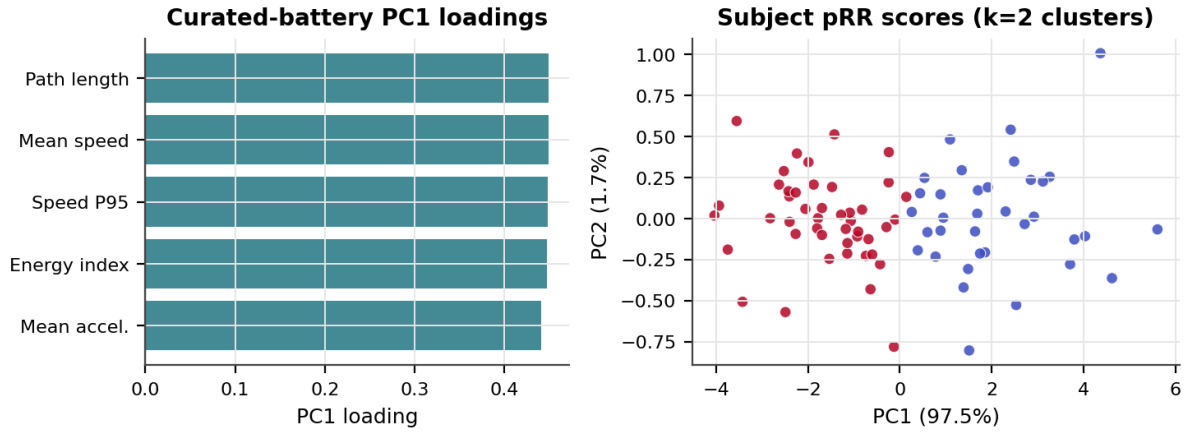


Figure 6: Latent structure of subject pRR profiles in the curated battery. Left: PC1 loadings (all positive and near-equal). Right: subject scores on PC1–PC2 coloured by $k = 2$ cluster assignment, showing clean separation along PC1.

Table 3: Internal consistency, dimensionality, and clustering across split-features ($N = 80$). α : Cronbach’s alpha; PC1: first-component explained variance; k : silhouette-selected clusters; sil: silhouette; ARI: adjusted Rand index of each estimator’s clustering versus D_{geo} .

Split feature	$\alpha(\text{all})$	PC1(all)	$\alpha(\text{cur})$	PC1(cur)	k	sil	ARI _A	ARI _B	ARI _C
ellipse_area95_m2	0.948	0.598	0.964	0.975	2	0.578	0.808	0.901	0.719
mean_speed_mps	0.946	0.564	0.960	0.960	2	0.505	0.854	1.000	0.557
path_length_m	0.946	0.564	0.960	0.960	2	0.505	0.854	1.000	0.557
rms_total_m	0.951	0.608	0.965	0.977	2	0.558	0.853	0.901	0.520
speed_p95_mps	0.949	0.569	0.960	0.960	2	0.501	0.854	1.000	0.719

Second, the subject-level pRR profile is internally consistent and, in a compact curated battery, near-unidimensional. Once an estimator is fixed, a single latent “pRR magnitude” captures most of the between-subject structure, and a two-profile partition is recovered robustly across estimators (except C). Practically, this supports reporting a small curated battery rather than an unfiltered bank of correlated measures. The energy-expenditure index behaves consistently with this picture: as a within-subject HIGH/LOW ratio it loads with the kinematic measures on the same dominant component, contributing to the shared sensory-balance dimension rather than introducing an independent axis. More generally, because pRR is a within-subject ratio, any influence that acts as an approximately common per-subject scale factor is suppressed—an attractive robustness property for individual profiling, though one that also implies a single composite index can mask information visible in the absolute measures.

Implications for standardization. These re-

sults carry a concrete recommendation for clinical and sports-medicine practice, where Romberg-type indices are routinely reported without a fixed computational convention. Because the estimator choice measurably changes the numeric value of the ratio and the dispersion of its agreement, indices computed under different conventions are not directly comparable across studies or clinics. We therefore advocate (i) computing the ratio on the log scale and aggregating paired trials geometrically, (ii) reporting the estimator and the trial-partitioning rule explicitly alongside the value, and (iii) summarizing balance with a small, pre-registered curated battery whose internal consistency has been verified, rather than an ad hoc mixture of correlated measures. The reliability analysis sharpens point (ii): because a single trialwise ratio is only fairly reliable ($\text{ICC} \approx 0.4$) while the multi-trial aggregate is good ($\alpha \approx 0.8$, $E\rho^2 = 0.76$), a defensible Romberg-type index should be built from several repeated trials, and the number of trials should be reported as part

of the protocol.

Limitations. The central limitation is interpretational: the proxy labelling does not correspond to EO/EC, so pRR cannot be read as a physiological Romberg ratio reflecting visual deprivation; it summarises within-subject contrast between low- and high-sway trials under a single condition. The cohort is a convenience sample of young adults with some anomalous metadata entries, and the energy index is a simplified kinematic proxy rather than a validated metabolic measurement (Grokhovsky and Kubryak, 2018). Finally, we deliberately avoided extra signal preprocessing; studies that apply filtering should report it explicitly.

Future work. Two extensions would strengthen the framework. The first is external validation of the curated battery against an annotated EO/EC dataset, establishing how the proxy contrast relates to the physiological Romberg ratio it does not claim to reproduce. The second is a hierarchical Bayesian treatment of the trialwise ratios, with subject-level effects drawn from a group distribution, which would propagate trial-level uncertainty into the subject profiles and complement the generalizability decomposition reported here; it would also allow partial pooling to stabilise the per-subject estimates that the fair single-trial ICC shows to be noisy.

Conclusion

We frame the proxy Romberg ratio (pRR) as a transparent repeated-trial contrast score and show that its value is estimator-dependent. Across proxy-labelling choices, geometric aggregation of paired trialwise ratios (D_{geo}) is a robust default that best matches the conventional ratio-of-means while keeping tight log-scale agreement; the bootstrap median is comparatively unstable. A compact curated battery yields near-unidimensional structure and two reproducible subject profiles. The energy-expenditure index, computed from the Grokhovsky–Kubryak integral posture-regulation definition, integrates into this structure as part of the shared sensory-balance dimension. We recommend that stabilometry studies state their ratio estimator explicitly, report agreement and uncertainty on the log scale, and include sensitivity checks for proxy labelling and for the exact definition of any composite index.

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[TO FILL] — e.g., conceptualization, methodology, software, formal analysis, data curation, writing, visualization.

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Appendix: full measure battery

The 24 trial-level measures are: path length; mean and 95th-percentile speed; peak speed; speed coefficient-of-variation; mean and peak acceleration; total RMS and per-axis RMS_X , RMS_Y ; range_X , range_Y ; robust $(P_{95} - P_5)$ ranges per axis; 95% confidence-ellipse area; spectral centroid and Welch median frequency of the speed signal; approximate entropy of speed; velocity zero-crossing rates per axis; and the three energetic quantities (total index A , average power P , and energy-per-metre V_{exp}) of Equation (5). The curated five-measure battery used for clustering and dimensionality analysis is `{mean_speed_mps, path_length_m, mean_accel_mps2, energy_A, speed_p95_mps}`.